

PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems

Drawing Sheets: 8 figures

FIG. 1 — System Architecture Block Diagram (8-Layer Loop)

FIG. 2 — Energy Balance Comparison

FIG. 3 — SNN Architecture (LIF Neuron Model)

FIG. 4 — Energy Harvesting Circuit

FIG. 5 — Activity-Dependent Energy Dynamics

FIG. 6 — Hardware Reference Design

FIG. 7 — Validation Results Summary

FIG. 8 — Energy-Bounded Recursive Control Architecture

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed January 31, 2026

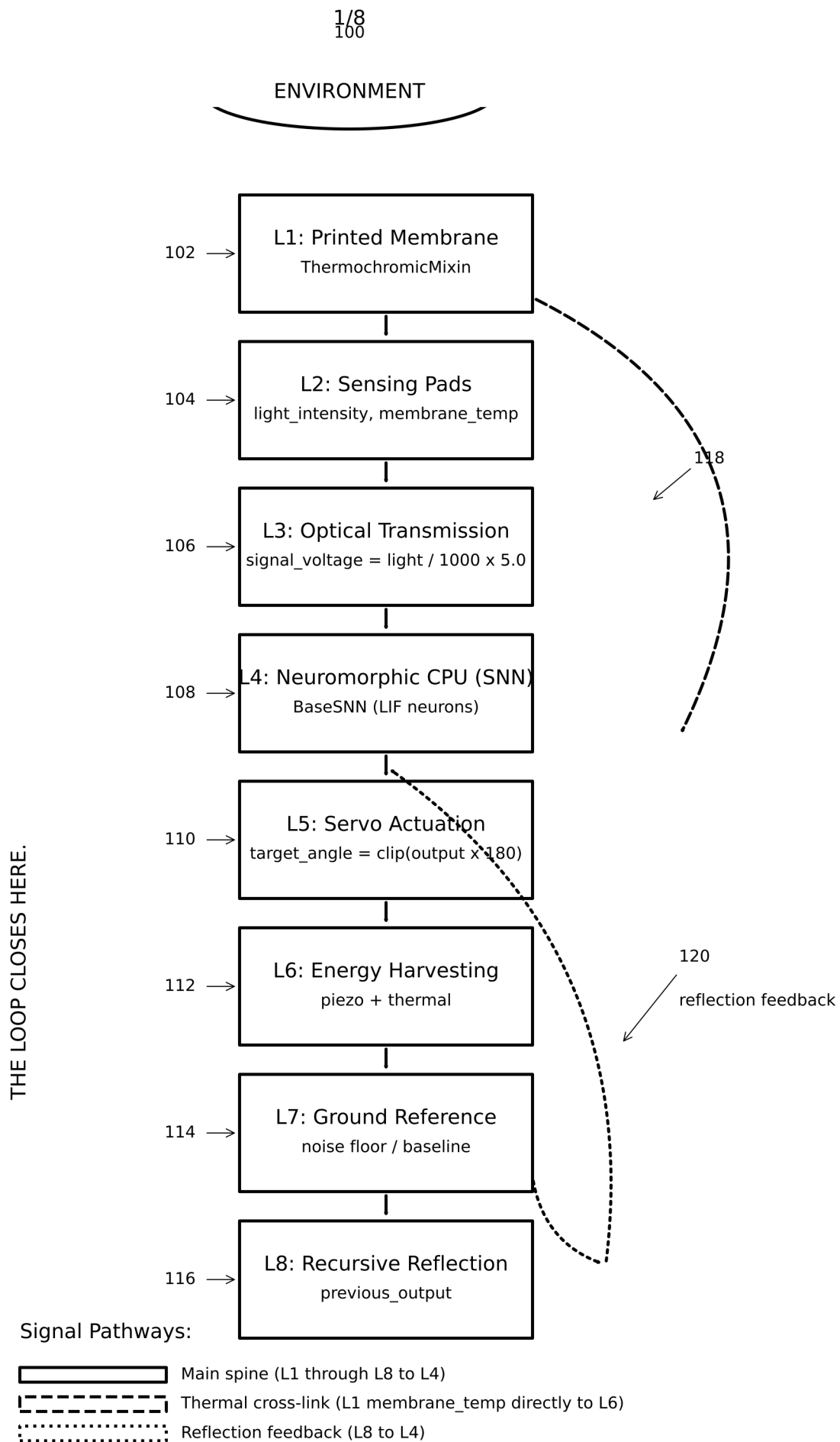


FIG. 1

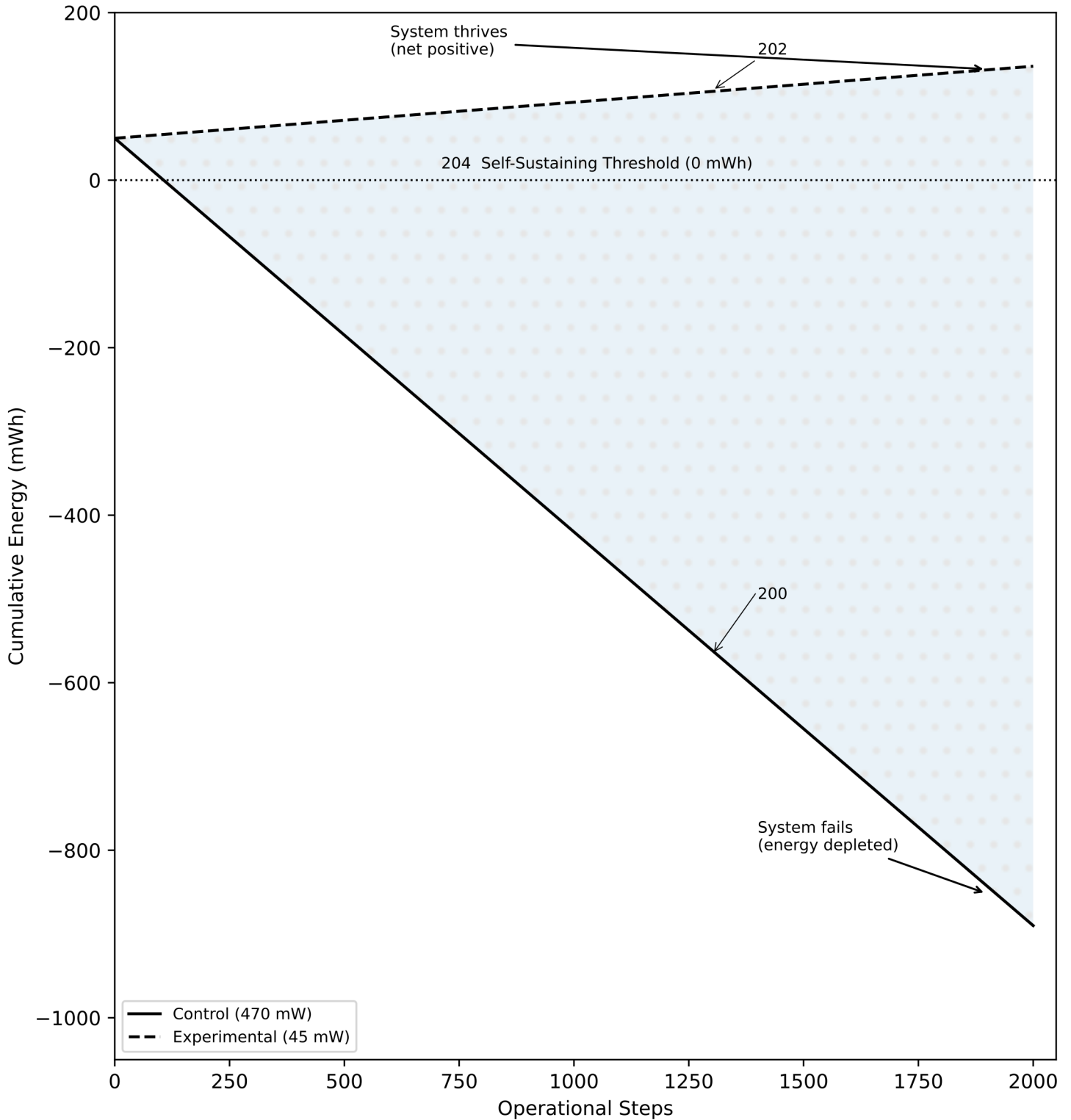


FIG. 2

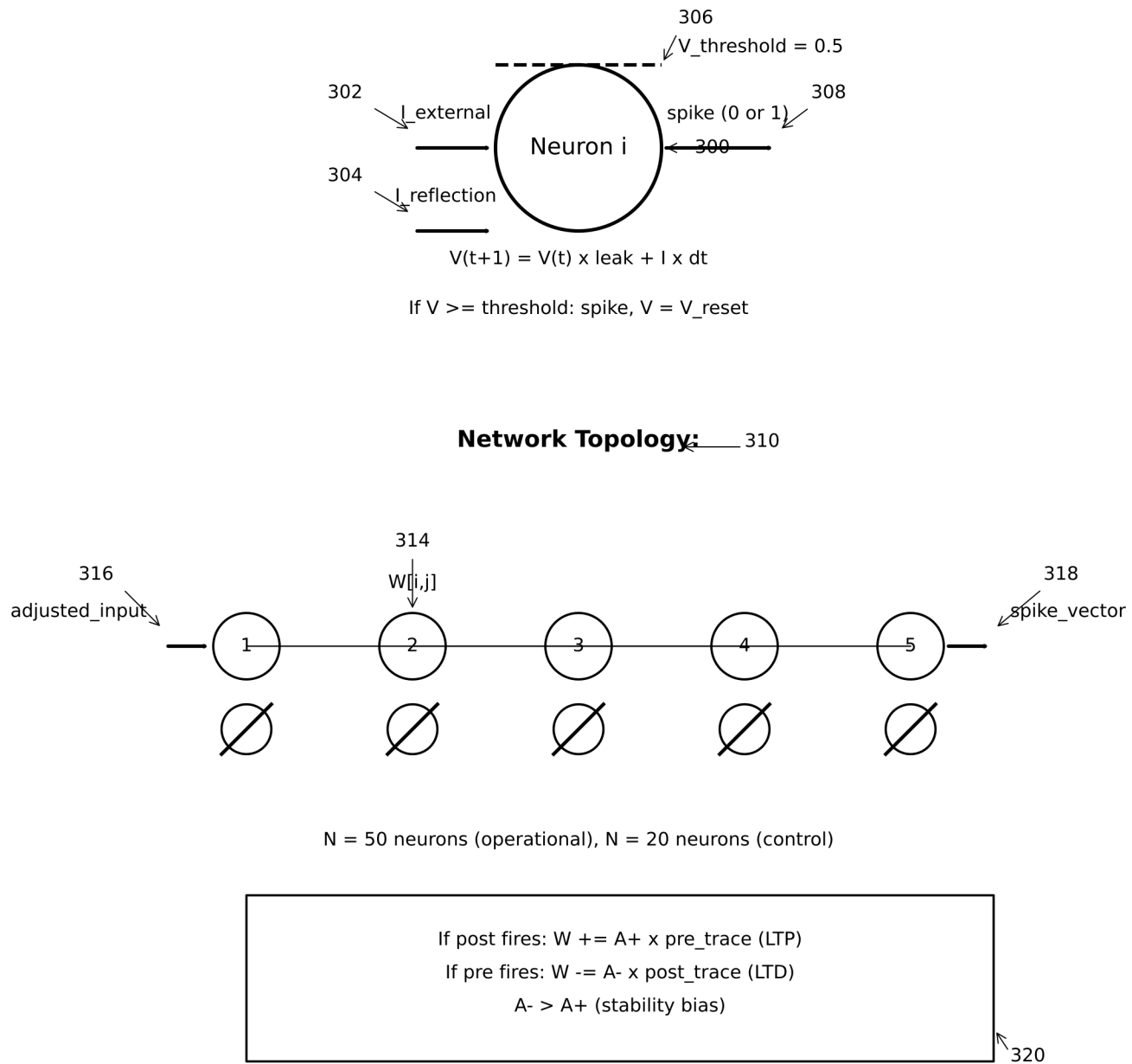


FIG. 3

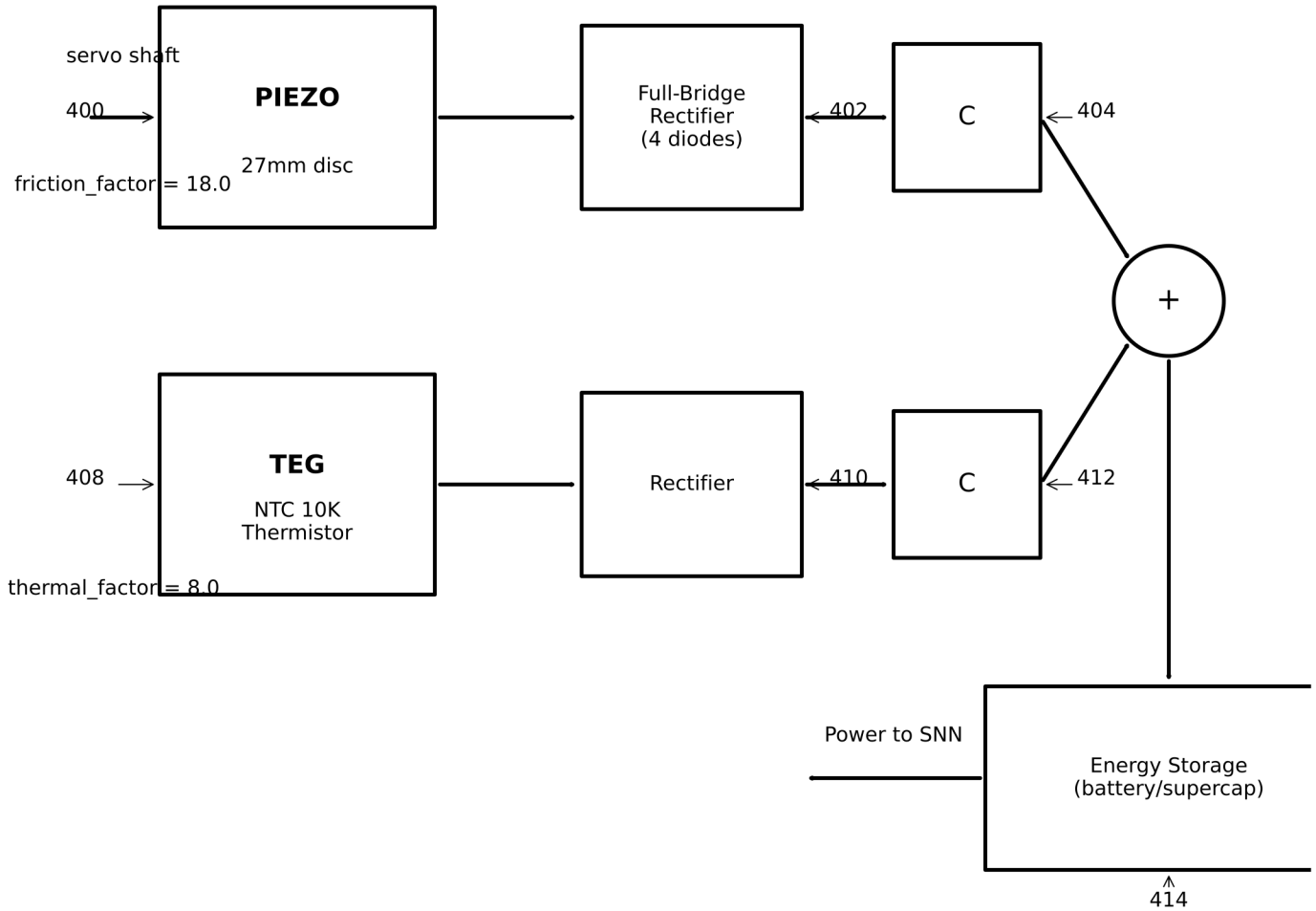


FIG. 4

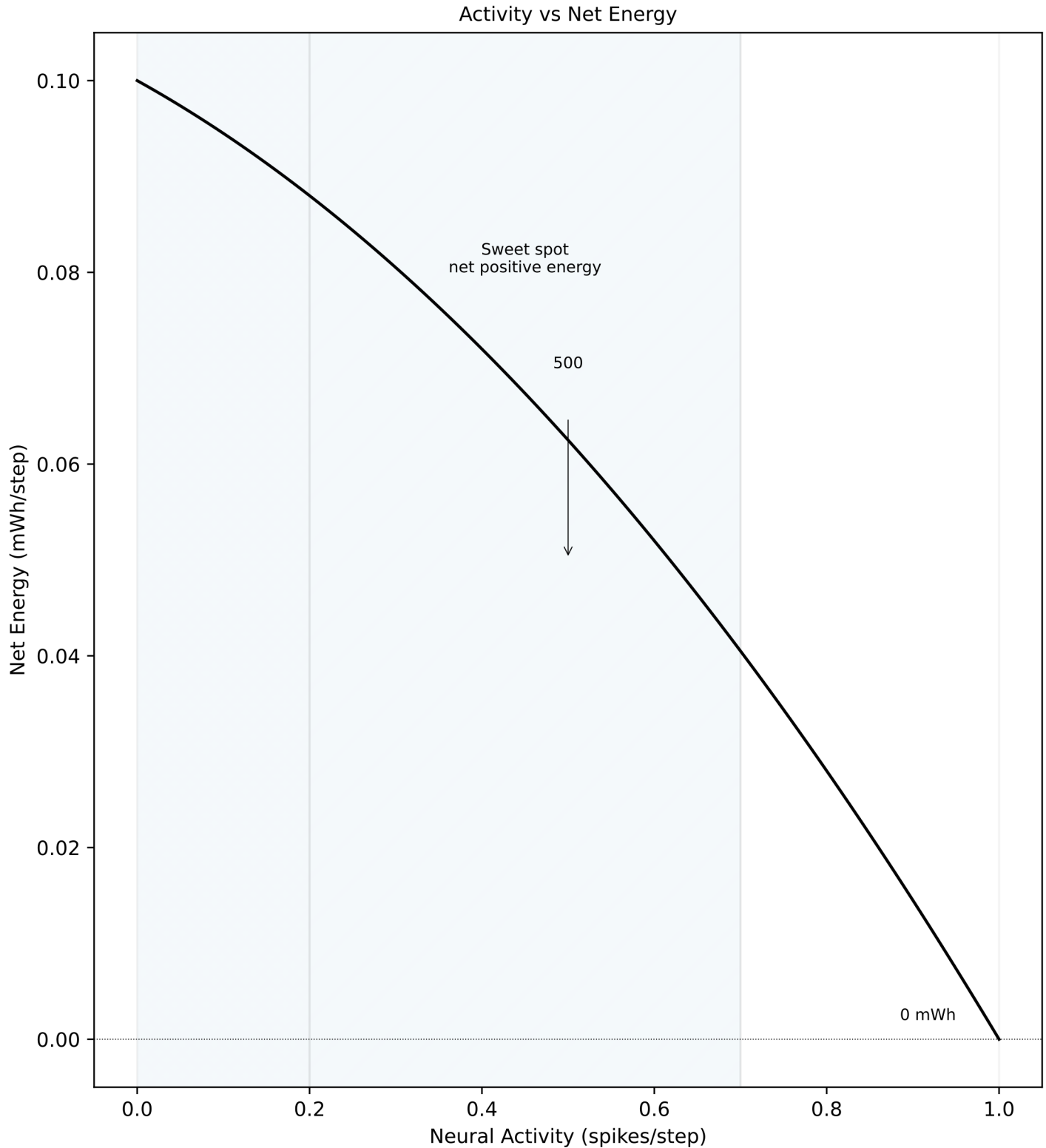
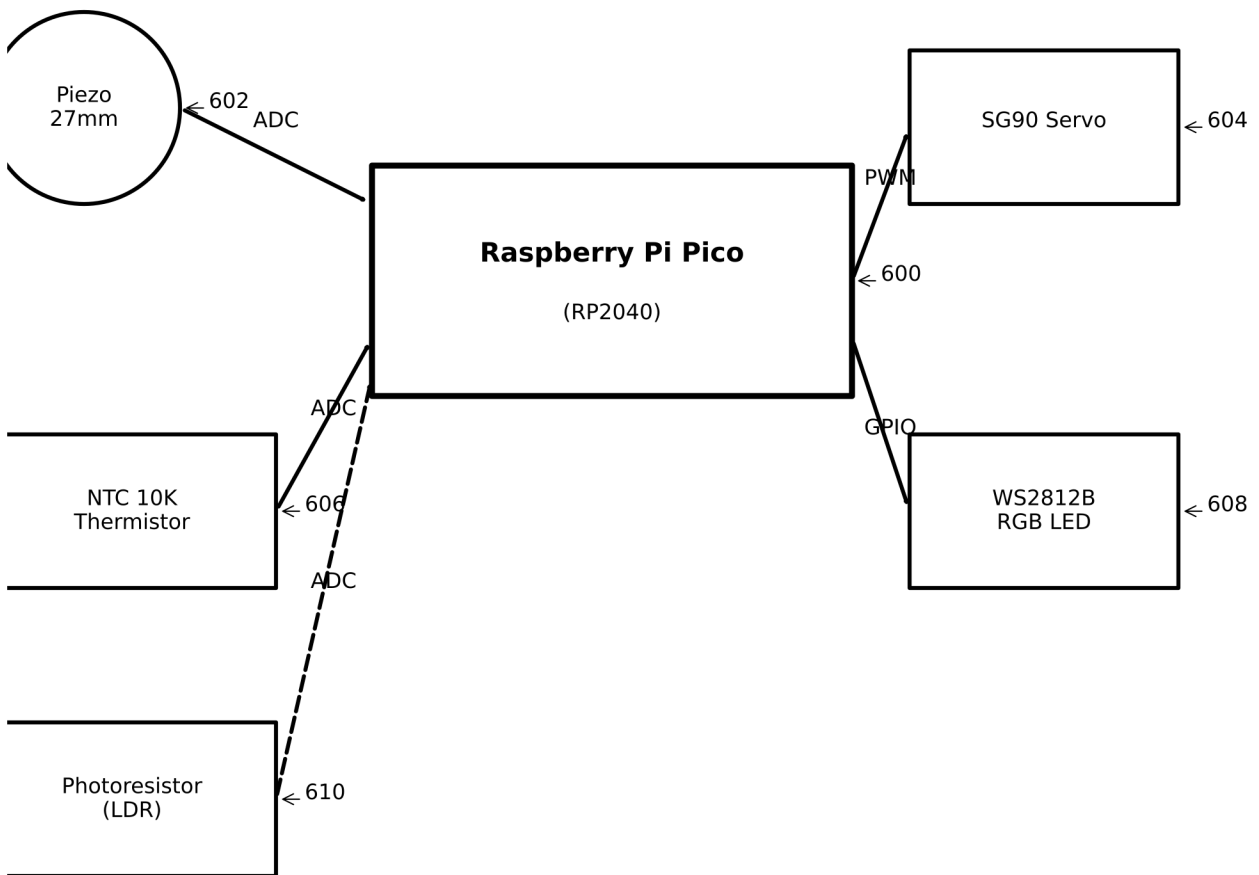


FIG. 5 Low activity
insufficient
harvesting

High activity
quadratic cost
dominates



Bill of Materials	
Raspberry Pi Pico	\$4
SG90 Micro Servo	\$3
27mm Piezo Disc	\$2
NTC 10K Thermistor	\$1
Photoresistor (LDR)	\$1
WS2812B RGB LED	\$1
Misc (wires, board)	\$3

Total BOM Cost: <\$25

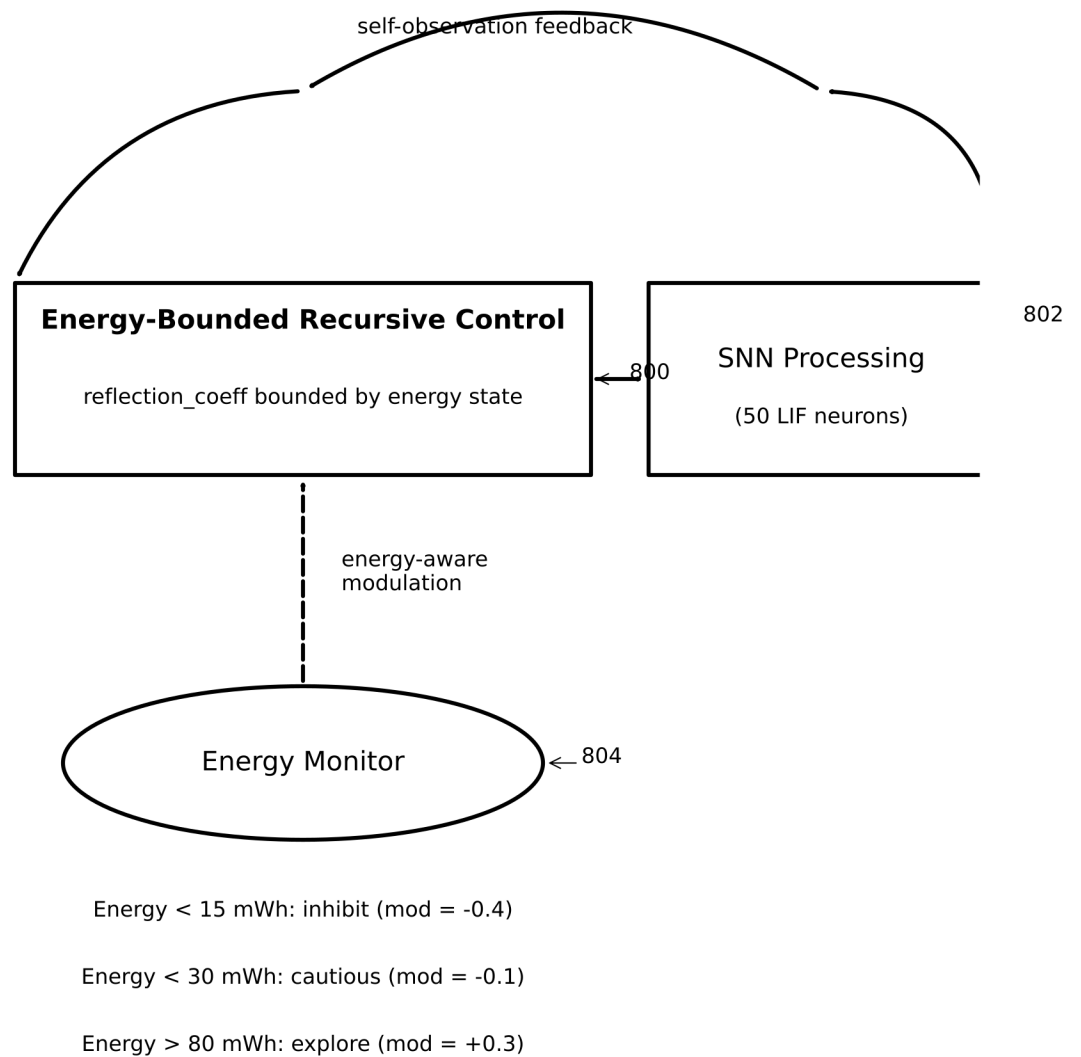
FIG. 6

Claim	Criterion	Measured	Threshold	Result
	Phase 7 Control Baseline (EnergyConfig)			
A	Closed-loop continuity	2000 pts	2000	PASS
B	Boundedness	bounded	bounded	PASS
C	Robustness (noise)	$\text{Std}(\theta) \leq 45^\circ$	$\leq 45^\circ$	PASS
D	Input-output gain	$\text{corr} \geq 0.2$	≥ 0.2	PASS
E	Saturation ratio	$\text{sats} \leq 0.20$	≤ 0.20	PASS
	Phase 7 Full Integration (Control)			
--	Energy at 2000 steps	-4.697 mWh	< 0	CONFIRMED
	MCP Integration (BalancedEnergyConfig)			
--	Energy at 2000 steps	+4.170 mWh	> 0	CONFIRMED
	Phase 8 STDP			
F8.1	Weight entropy decrease	2.95~2.02	decrease	PASS
F8.2	STDP MI > 1.2x frozen	6.52x	> 1.20	PASS
F8.3	Weight convergence	0.00%	$< 10\%$	PASS

700
v

All claims validated. System reproducible via git tags.

FIG. 7



Recursive control loop bounded by energy levels to ensure self-sustainability.

FIG. 8

PATENT B

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

Drawing Sheets: 6 figures

FIG. 1 — Self-Observation Feedback Loop

FIG. 2 — Reflection Coefficient Spectrum

FIG. 3 — Dynamic Modulation Sources

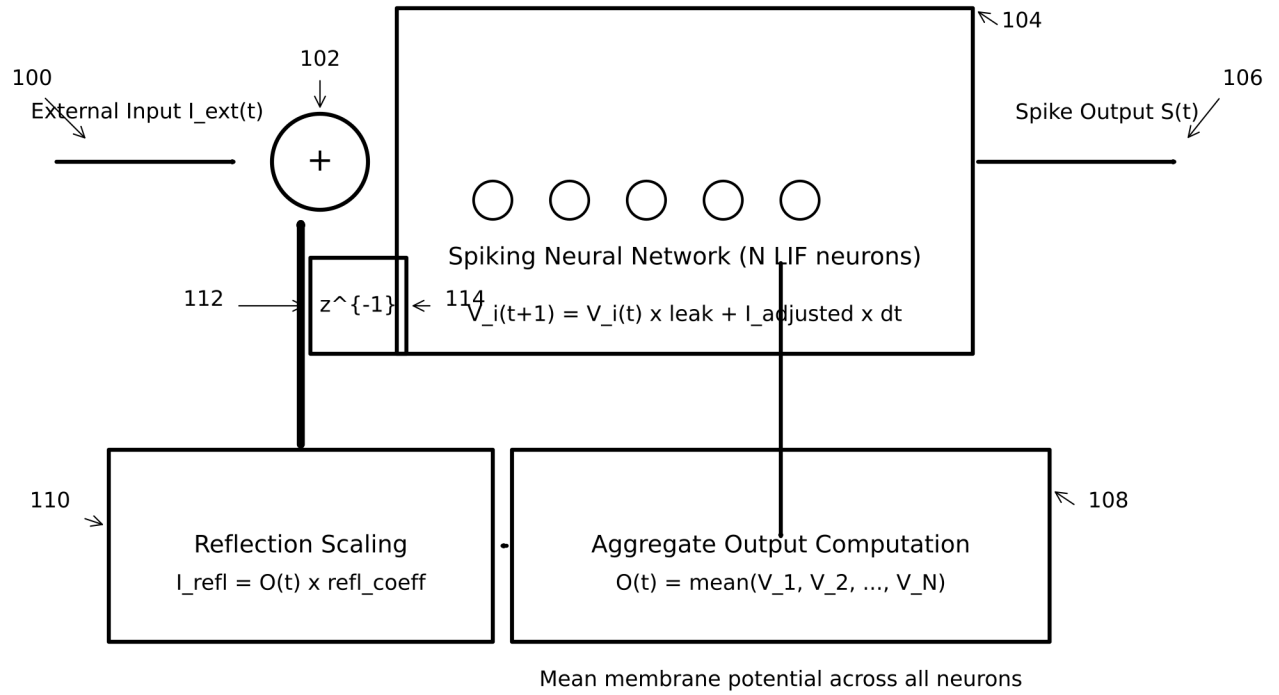
FIG. 4 — Self-Referential Learning Loop (STDP + Self-Observation)

FIG. 5 — Energy-Aware Self-Observation Regulation

FIG. 6 — End-to-End Signal Flow with Self-Observation Integration

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed January 31, 2026



Self-observation: the network observes its own prior aggregate state

$$I_{\text{adjusted}}(t+1) = I_{\text{ext}}(t+1) + I_{\text{reflection}}(t)$$

FIG. 1

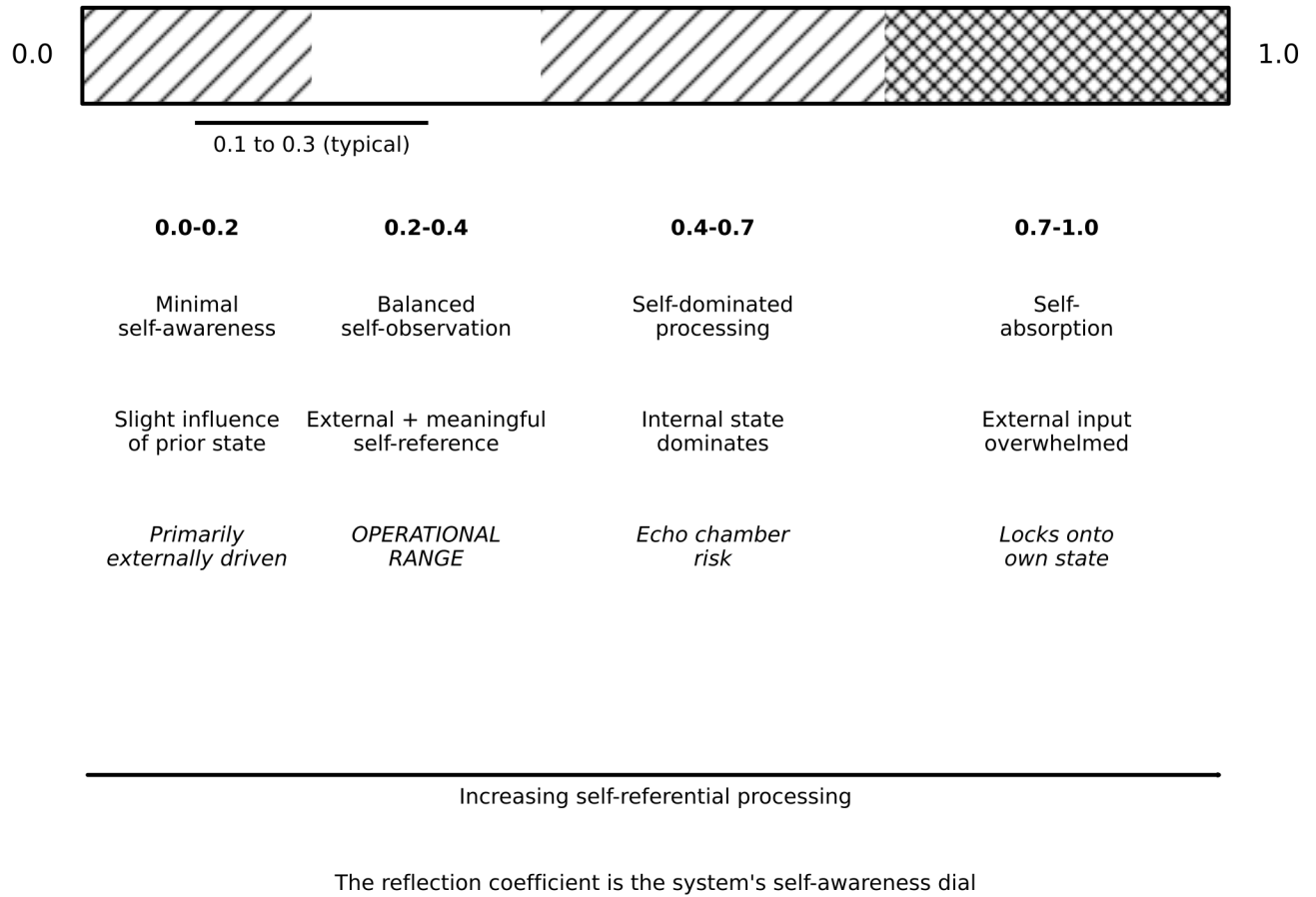


FIG. 2

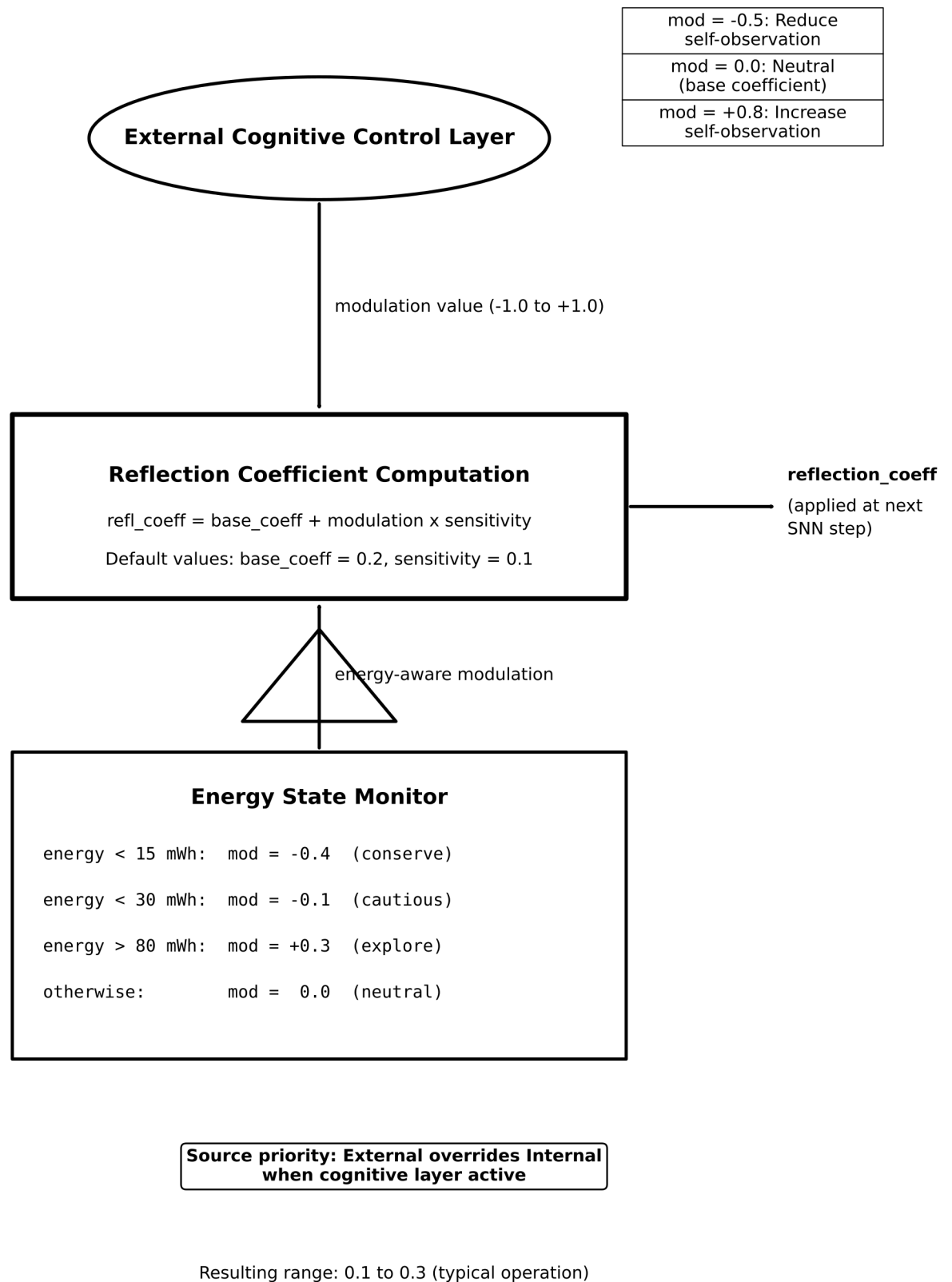


FIG. 3

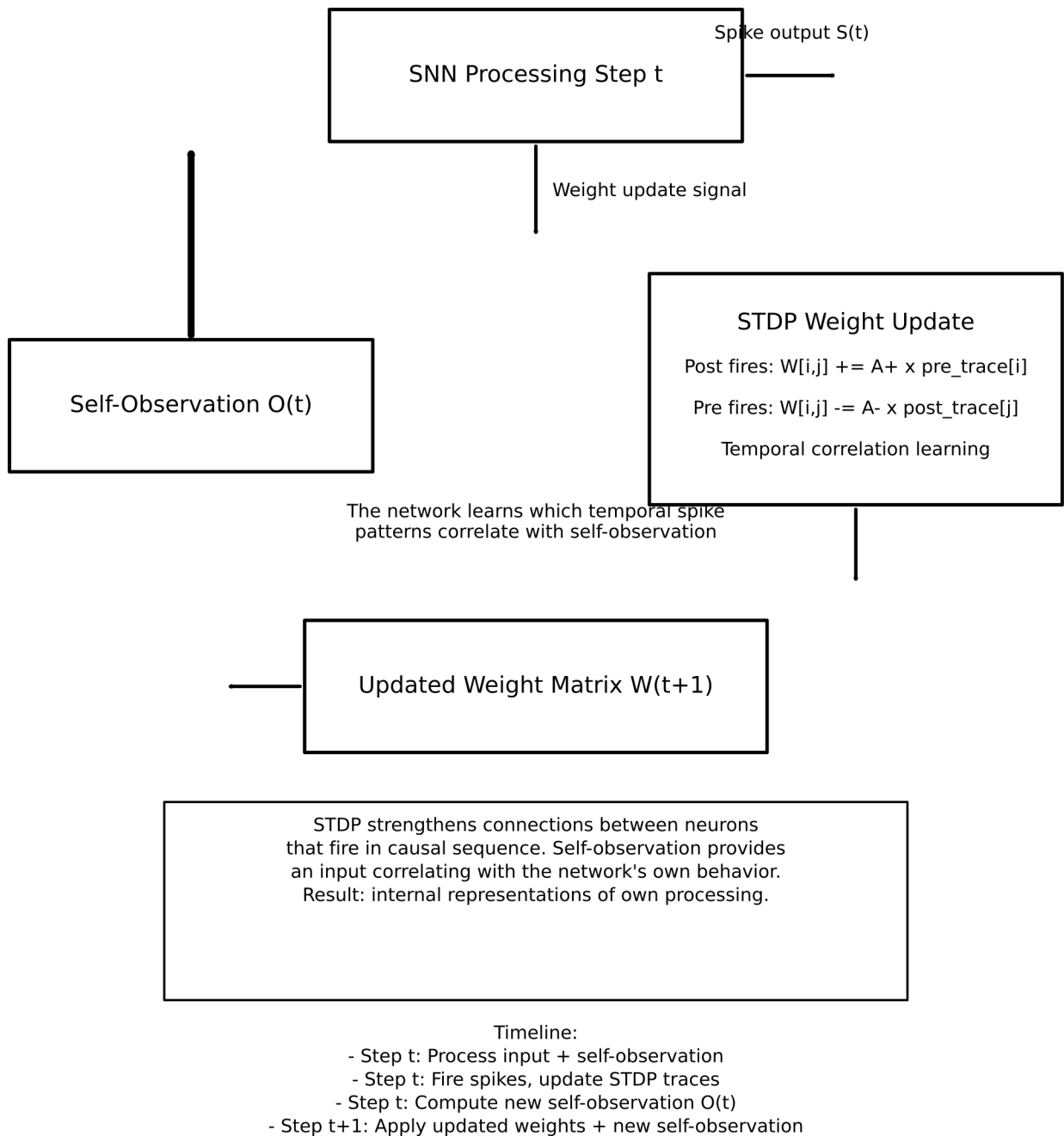
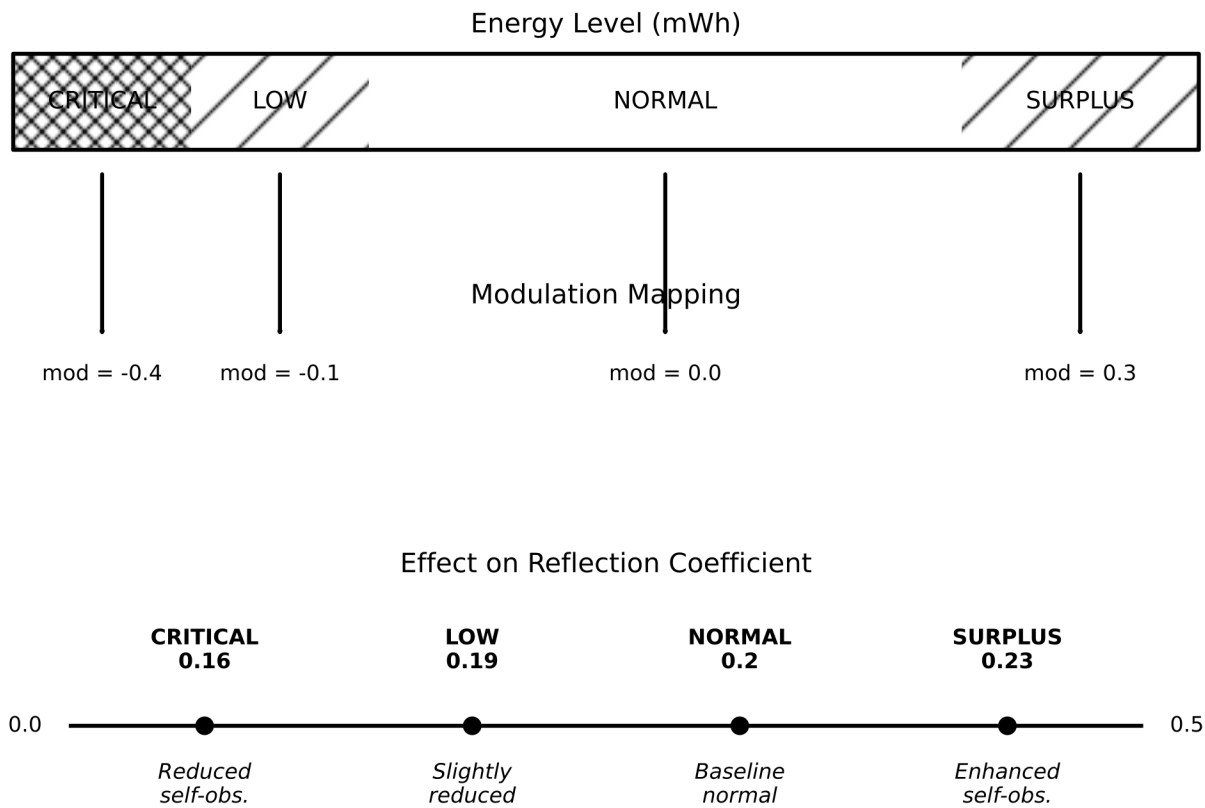


FIG. 4



Energy-aware regulation ensures the system reduces cognitive overhead when energy is scarce and explores deeper self-reference when energy is abundant

FIG. 5

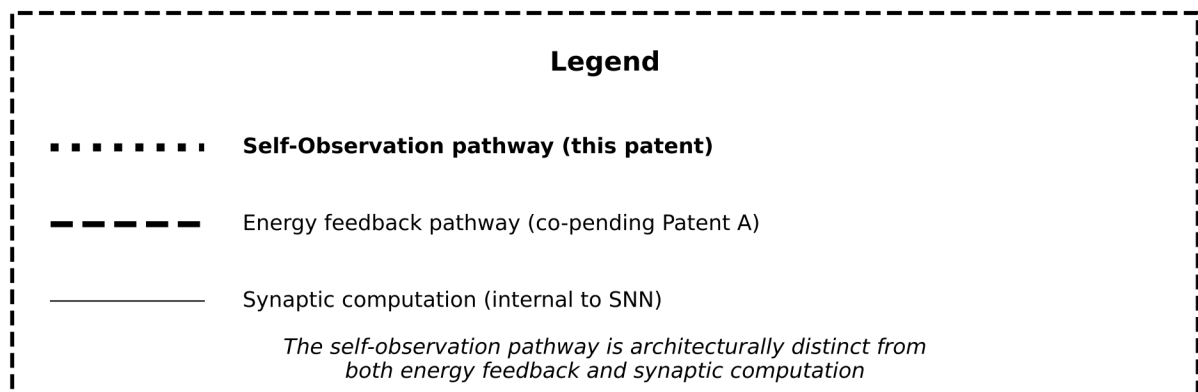
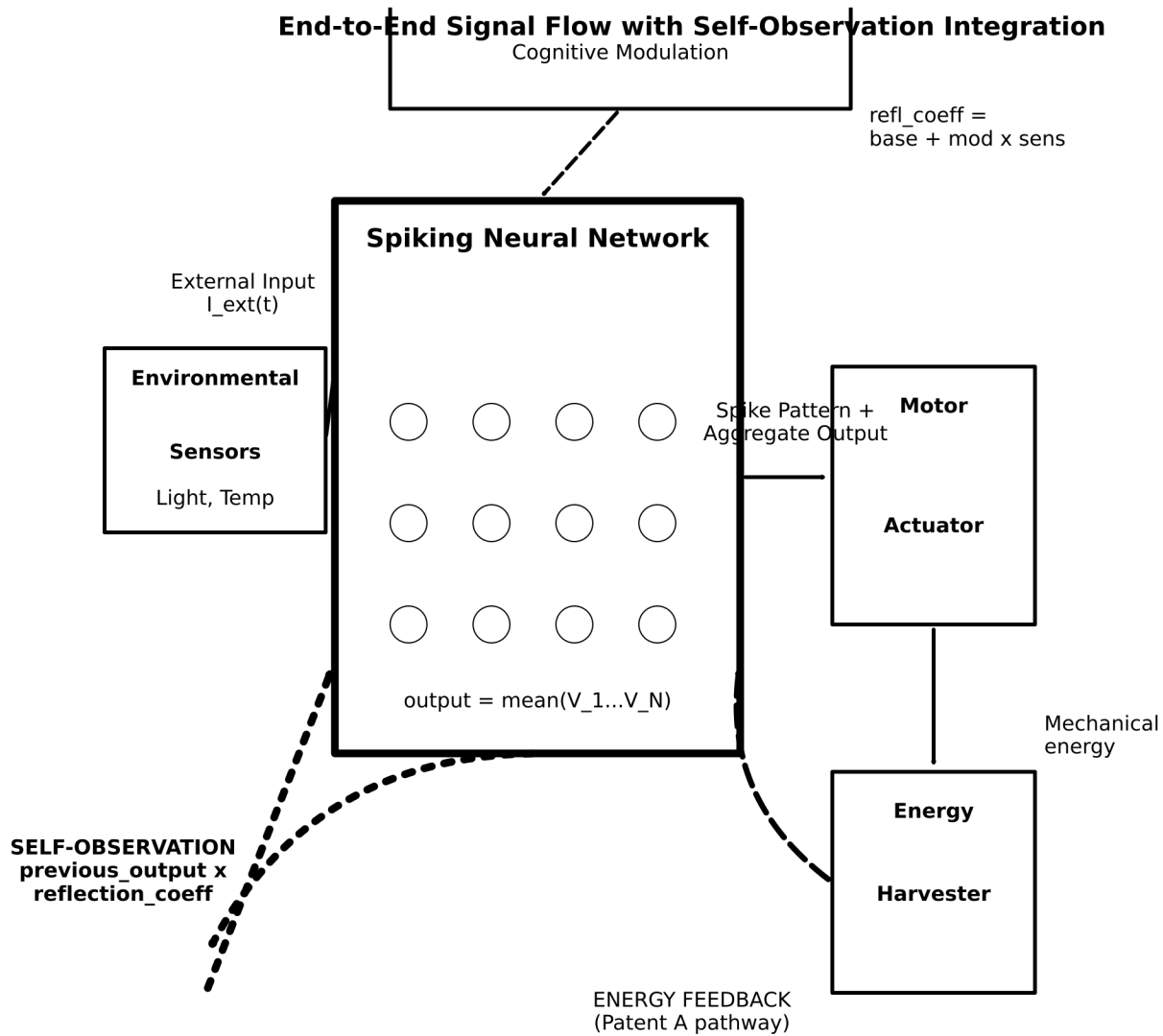


FIG. 6

PATENT C

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

Drawing Sheets: 7 figures

FIG. 1 — System Architecture with Fallback

FIG. 2 — State Transition Diagram

FIG. 3 — Energy-Aware Modulation Curve

FIG. 4 — Autonomous Input Generator Output

FIG. 5 — Resynchronization Payload Structure

FIG. 6 — Recovery Timeline Diagrams

FIG. 7 — End-to-End Signal Flow: Connected vs. Autonomous

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed January 31, 2026

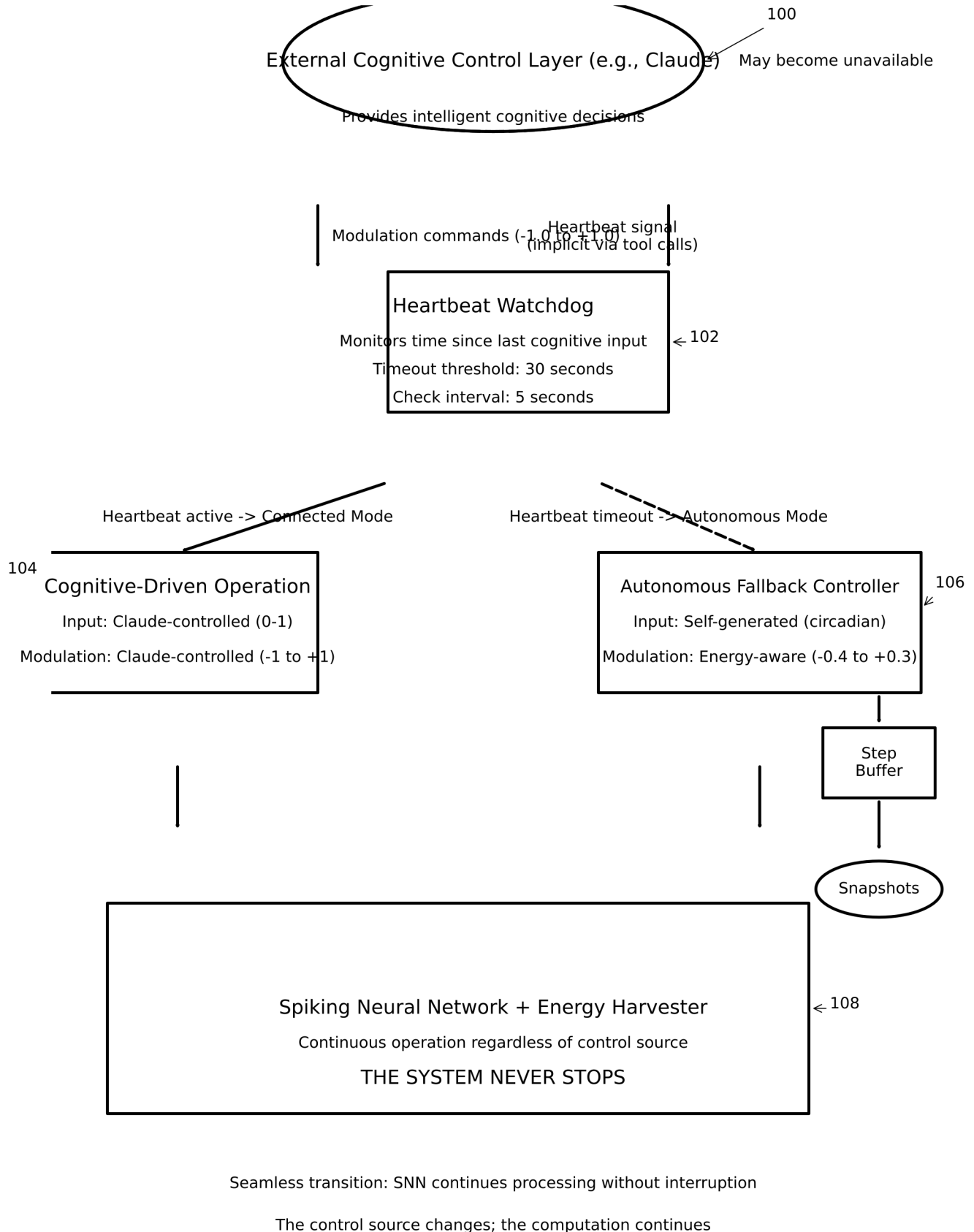


FIG. 1

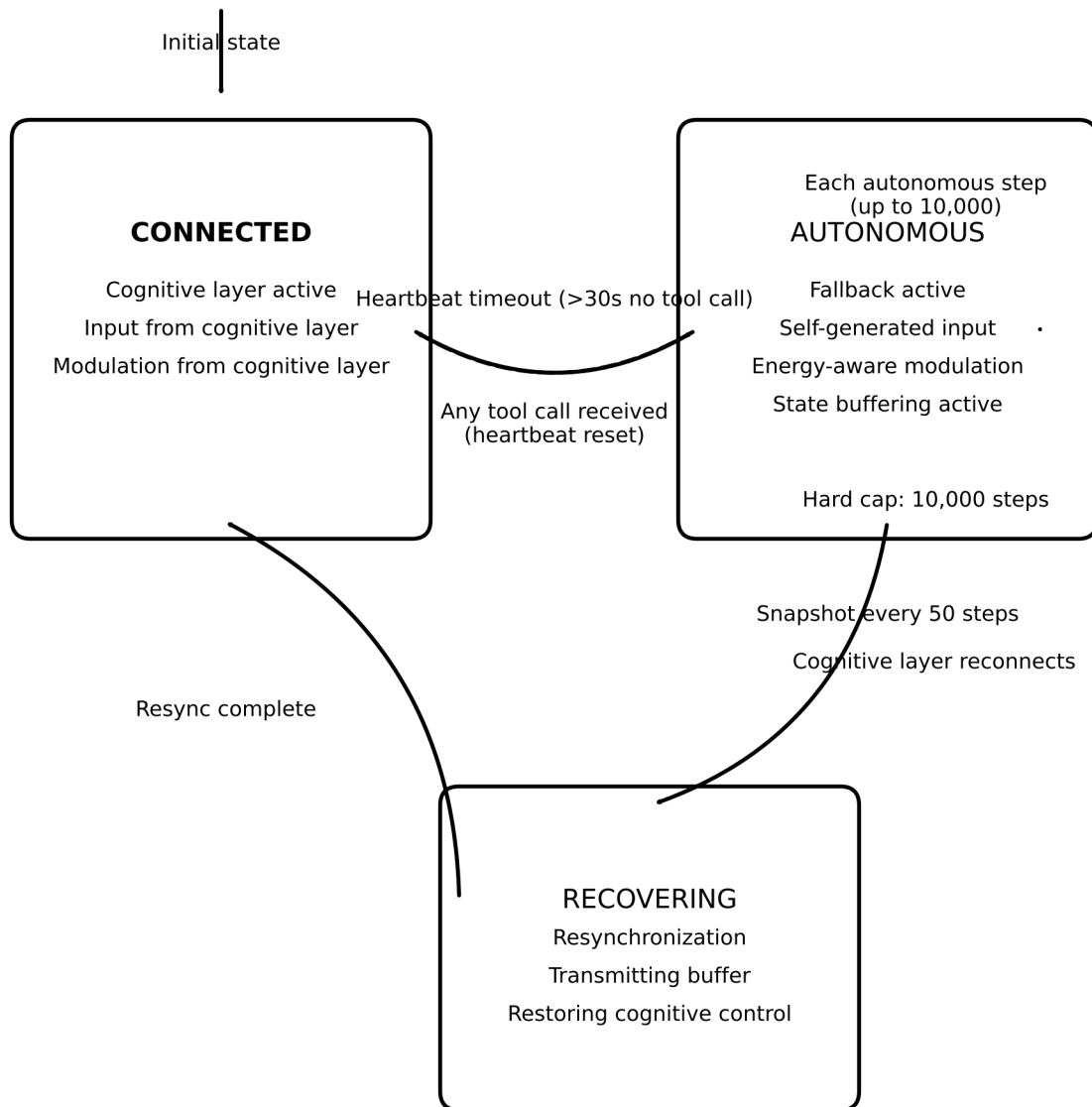
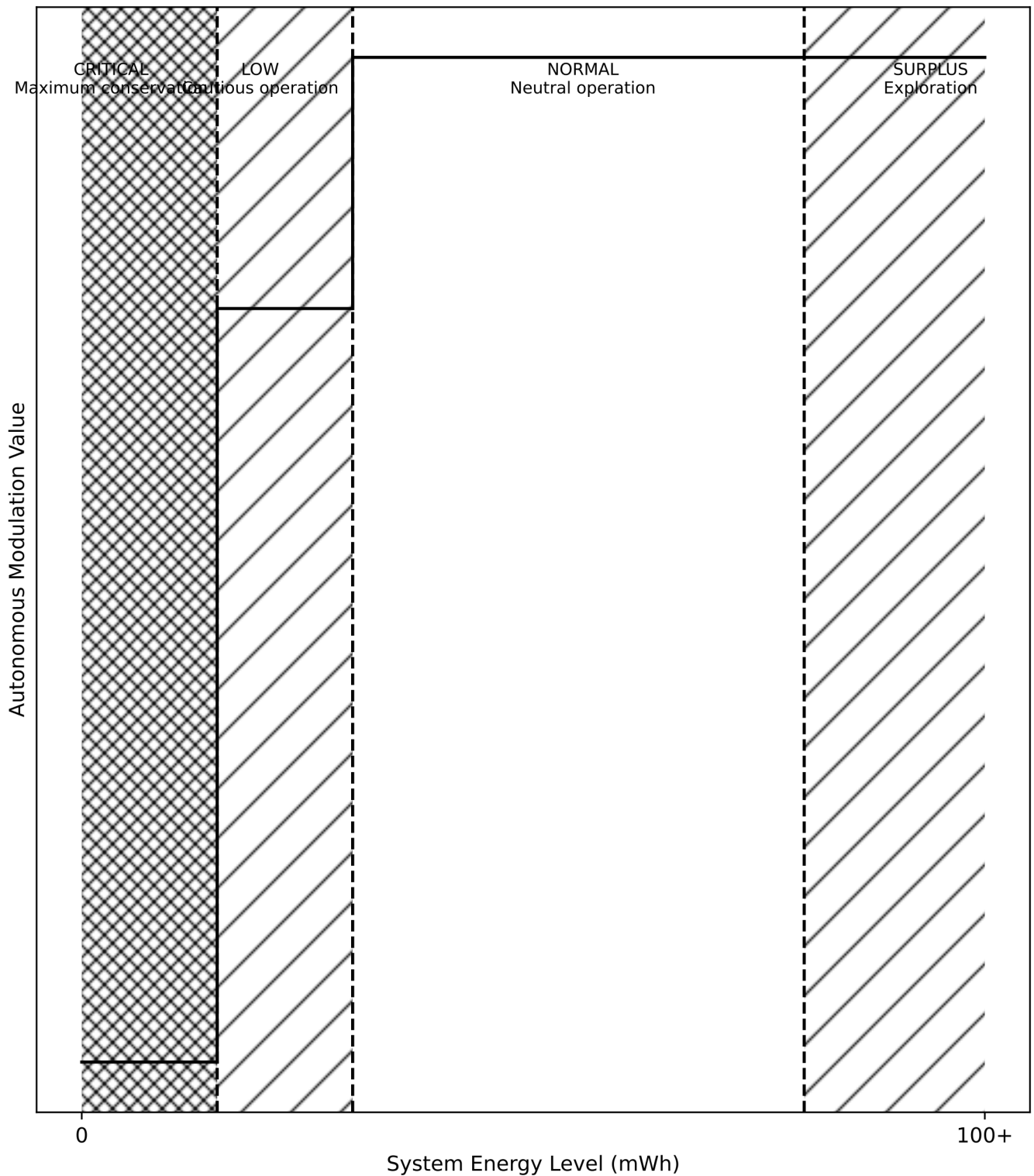


FIG. 2



System autonomously adjusts processing intensity based on energy — mimicking metabolic regulation

FIG. 3

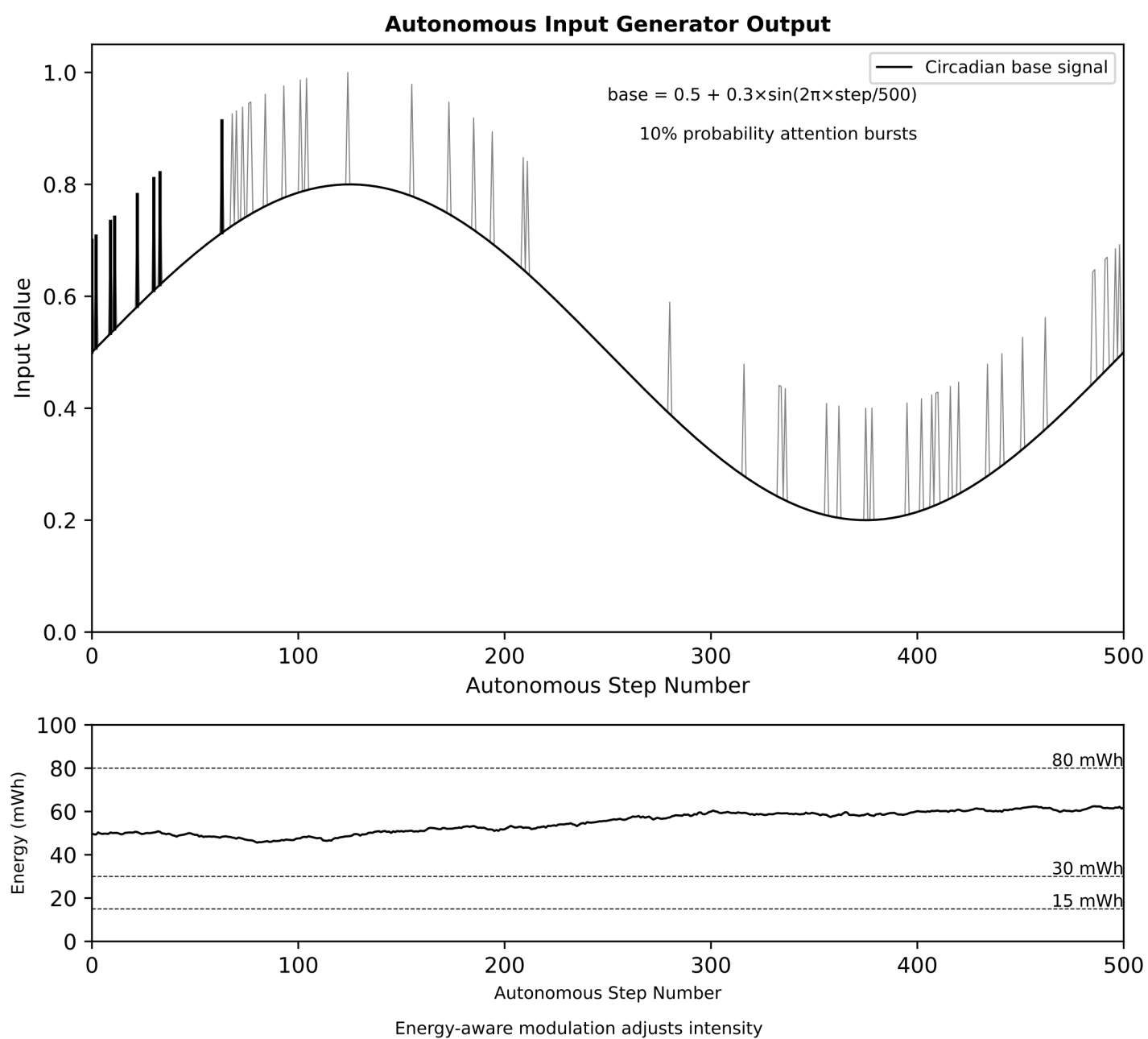


FIG. 4

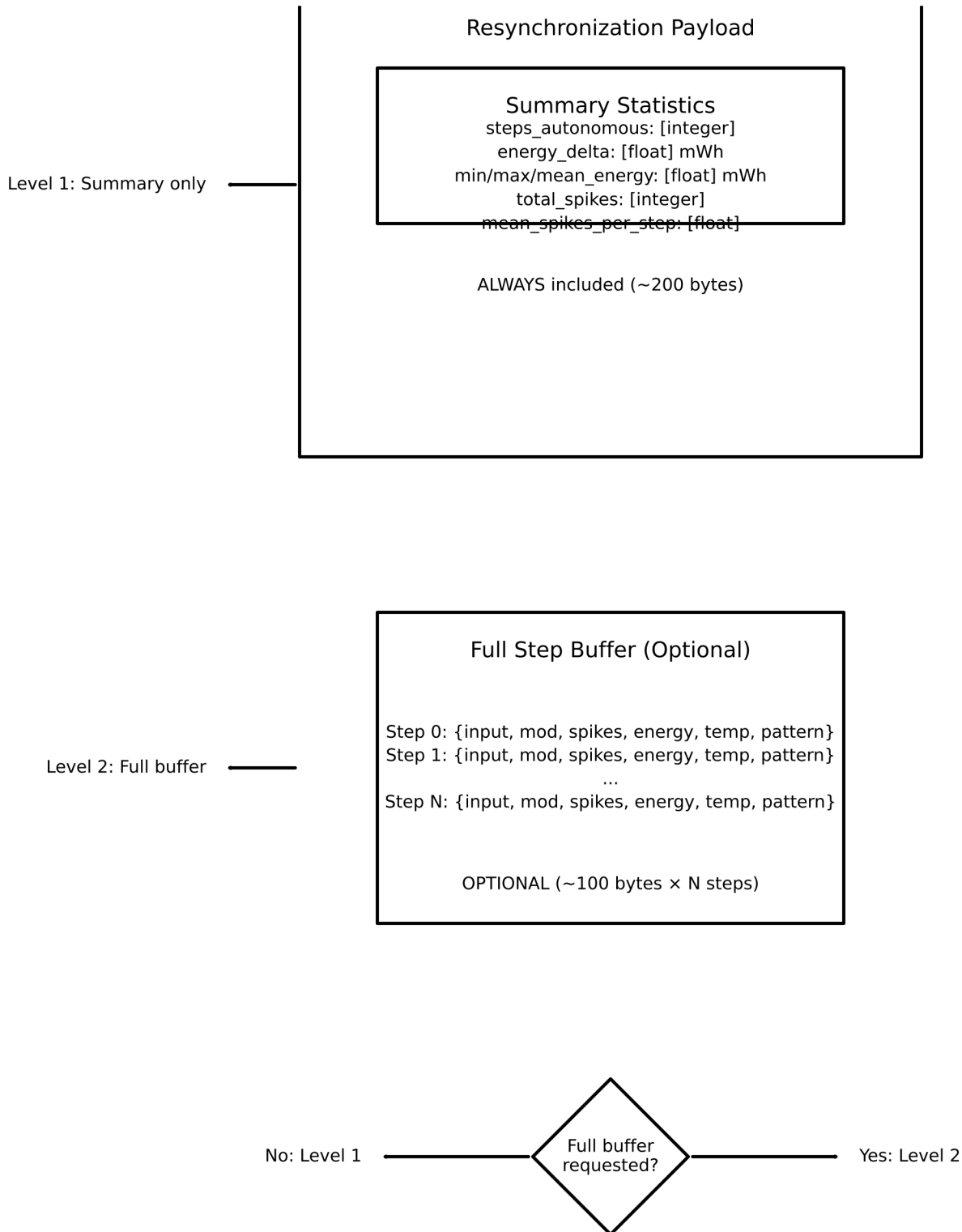
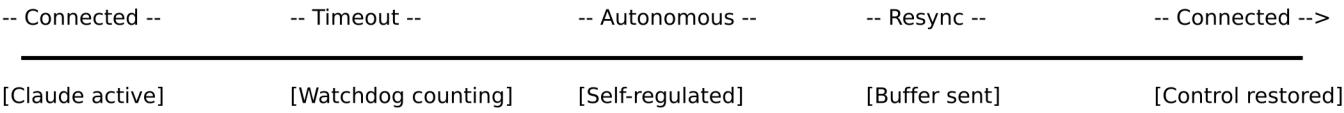
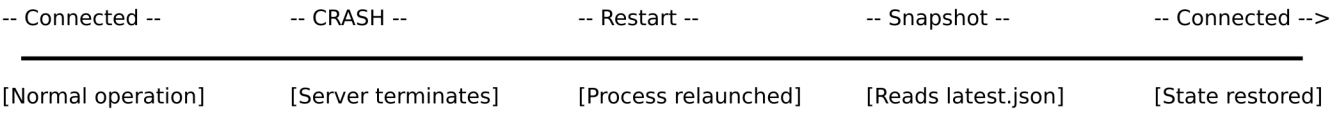


FIG. 5

Timeline A — Normal Reconnection:

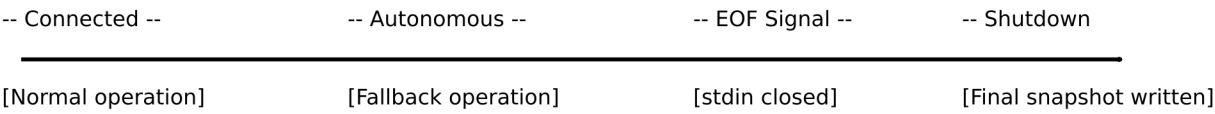


Timeline B — Crash Recovery:



Max state loss = 50 steps (snapshot interval)

Timeline C — Clean Shutdown:



State preserved for next session — zero data loss

Solid segments: System actively processing

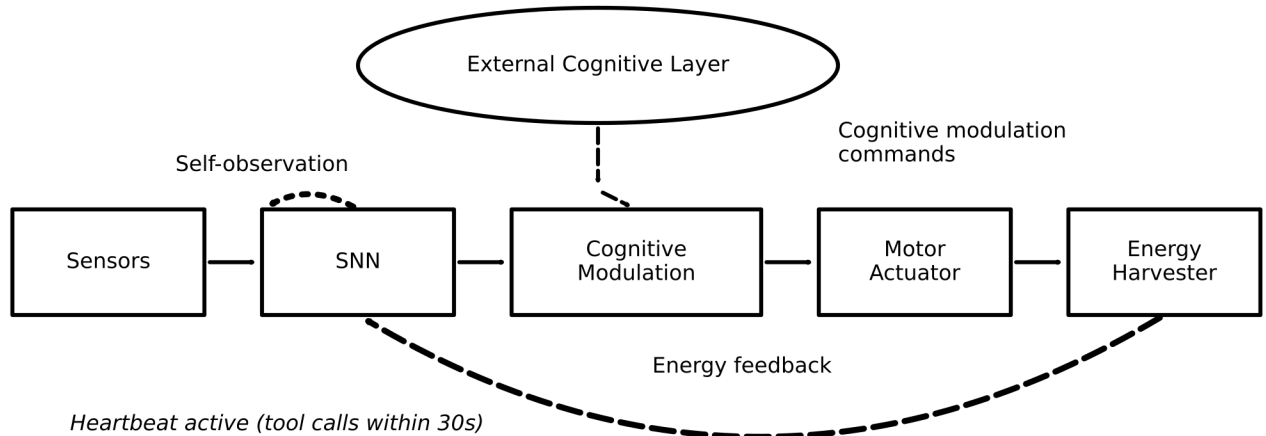
Dashed segments: System in transition

Bold marker at each state change: No data loss at any transition

FIG. 6

End-to-End Signal Flow: Connected vs. Autonomous Operation

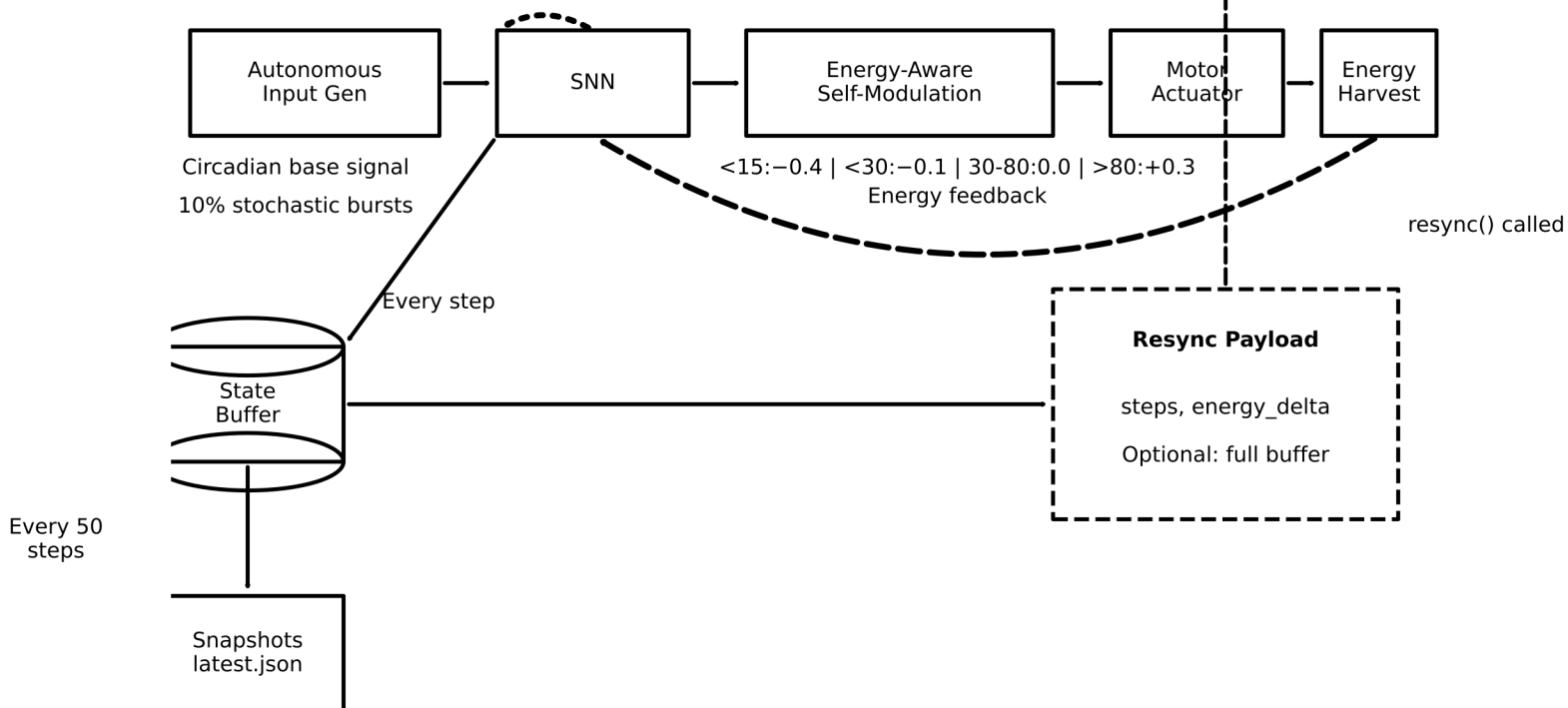
CONNECTED MODE



DISCONNECTION EVENT (heartbeat timeout > 30s)

Seamless transition — no interruption to SNN or Harvester

AUTONOMOUS MODE



The fallback system changes the INPUT SOURCE and MODULATION SOURCE

The core neural processing, motor actuation, and energy harvesting

continue **WITHOUT INTERRUPTION** through the transition

FIG. 7

— Forward signal - - - - - Energy feedback Self-observation